

Problem 4.22

If $A(4) = \$1,200$ and $A(n) = \$1,800$,

(a) what is $A(0)$, assuming a simple interest of 6%?

(b) what is n if $i = 0.06$ assuming simple interest?

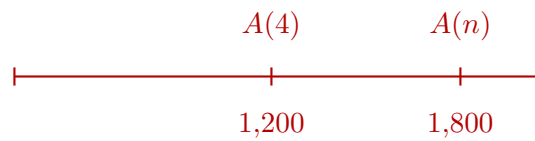
Solution

Under simple interest,

$$A(t) = P(1 + it).$$

Here,

$$i = 0.06.$$



For part (a),

$$1200 = P(1 + 0.06 \cdot 4) = P(1.24).$$

Therefore,

$$P = \frac{1200}{1.24} = 967.7419.$$

So,

$$\boxed{A(0) = \$967.74}.$$

For part (b), use $A(n) = 1800$:

$$1800 = 967.7419(1 + 0.06n).$$

Hence,

$$\frac{1800}{967.7419} = 1 + 0.06n.$$

So,

$$\left(\frac{1800}{967.7419} - 1 \right) \div 0.06 = n.$$

Thus,

$$n \approx 14.33.$$

Therefore,

$$\boxed{n \approx 14.33}.$$